7

COST AND REVENUE

QUICK REVISION

Concepts of Cost

The following concepts of costs are used in price theory:

1. Money Cost: Money costs include all expenses incurred on factor-inputs engaged in the production of a commodity. These costs are also called accounting costs or explicit costs.

2. Implicit Costs and Explicit Costs: Implicit costs are the costs of self-owned resources while all cash payments made by the firm to outsiders for their services and goods are known as explicit costs. The sum of explicit and implicit costs is total cost of production.

3. Economic cost = Explicit cost + Implicit cost + Normal profit.

4. Normal Profit: The minimum amount which is required to keep entrepreneur in the production is known as non profit. This normal profit is in a way reward or remuneration for an entrepreneur and therefore should be treated as cost.

5. Real Cost: Real cost is expressed not in money terms but in terms of efforts of labour and sacrifices by the producers in producing the commodity. It is more psychological concept and cannot be measured and it is not applied actual practice.

6. Private Cost: Private cost refers to cost of production incurred by an individual firm in producing a commodity Private cost is same as economic cost.

7. Social Cost: Social cost on the other hand refers to the cost that the society has to bear on account of production a commodity.

Social cost = Private cost + External cost

8. Opportunity Cost: Opportunity cost (also known as alternative cost) of a commodity is the value of the sacrificed product which could have been produced by using the same resources.

The concept of opportunity cost is very useful in economics. It is applicable to the pricing of any commodity factor of production. It is also useful to consumers in their decision making.

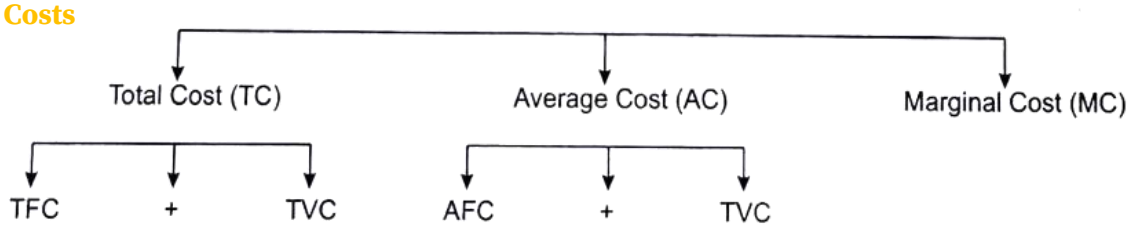
9. Fixed Cost: Fixed costs are those costs of production which are incurred on fixed inputs. They remain fixed levels of output. They are also called as supplementary costs or overhead costs and unavailable cost.

10. Variable Cost: Variable costs vary directly with the level of output. So, they are also known as direct costs. are zero at zero level of output. TVC curve starts from the point of origin and moves upwards to the Initially, it rises at a diminishing rate, then at a constant rate and ultimately at an increasing rate.

11. Cost Function: The functional relationship between cost and quantity produced is termed as cost function,

$$C = F(Q_X)$$

Short Run Costs



1. Total Fixed Cost: Fixed costs are the costs which do not vary with the output. These are the expenses incurred on the fixed factors of production. These are also known as supplementary costs or overhead or unavoidable costs. that total fixed cost remains the same at all TFC curve is a horizontal line parallel to the x-axis which tells us that levels of output.

2. Total Variable Cost: Variable costs are the costs which vary directly with the output. These are the expenses incurred on the variable factors of production. They are also referred to as prime costs or special costs or direct costs or avoidable costs.

TVC curve starts from the origin because at O output it is zero. TVC curve is first concave downward indicating TVC increases at decreasing rate and after a certain point it becomes concave upward which indicates that TVC increases at increase rate.

The shape of TVC follows directly from the law of variable proportions. It diminishes due to increasing returns to the variable inputs arising from fuller utilisation of fixed factors and greater specialisation. It increases due to diminishing returns to the variable inputs arising from difficulty of management and over utilisation fixed factors.

3.Total Cost: Total cost is the sum total of total fixed costs and total variable costs.

Since, TFC remains constant the changes in total costs are entirely due to changes in total variable cost TC & TVC curves have the similar shapes. The only difference is that TVC curves starts from origin while TC curves starts from above the origin

4. Average Fixed Cost: AFC is the per unit fixed cost of production. It can be obtained by dividing TFC by quantity of output. In case of AFC as the output rises the curve goes on declining. The AFC curve is a downward sloping curve as output approaches to zero. AFC approaches to infinity but it never touches Y-axis. On the other hand, when output approaches at very high level AFC approaches zero but it will never become zero it will run parallel to X-axis. Thus, AFC takes the shape of rectangular hyperbola.

5.Average Variable Cost: It is the per unit variable cost of production of a commodities. It can be obtained by dividing TVC by the quantities of output. Thus, AVC = TVC/Output.

As output rises the AVC curve 1st falls reaches a minimum and then begins to rise. Thus, AVC has a U-shape. Its U-shape is because of returns to a factor or law of variable proportion upto a minimum point of optimum capacity the plant operates in the raw of increasing returns which reduces the average variable cost with the rise in output after the point of optimum capacity operates the law diminishing returns which increases the variable cost. Thus, AVC becomes a U-shaped arise curve.

6. ATC/AC: ATC/AC is obtained by dividing the TC with the quantities of output. It is per unit cost of production.

7. Marginal Cost: Additional cost of producing an additional unit of output is known as MC.

8. Revenue: The money receipts from the sale of the product is known as 'revenue'. These are of following three types.

(i) **Total Revenue**: It is the sum of all sale receipts or income of the firm. Thus, TR = Price X output.

(ii) **Average Revenue**: It is per unit revenue received from sale of a commodity, i.e., TR/Q. Average revenue is equal to price.

(iii) **Marginal Revenue**: It is the additional revenue earned from the sale of 1 additional unit of commodity, i.e.,

$$MR = TRn - T.Rn\text{-}1$$

1 MARK QUESTIONS

A. FILL IN THE BLANKS

1. Cost incurred on hiring outside factors of production is _____ cost.
2. Cost of owner's own provided factors of production is _____ cost.
3. Two **Example**s of explicit cost are _____ and _____.
4. Two **Example**s of implicit cost are _____ and _____.
5. Cost incurred only during the process of production is called _____.
6. Cost incurred even at zero level of production is _____ cost.
7. _____ is a horizontal cost curve.
8. _____ curve is an inverted S shaped curve.
9. AFC is shaped as a _____.
10. Rising portion of _____ curve is called supply curve of a competitive firm.
11. AC is a U-shaped curve due to _____.
12. When MC > AC, AC_____.
13. When MC<AVC, AVC_____.
14. When MC = AC, AC is at its_____.
15. As more units of a commodity are produced, AFC falls at a _____ rate.
16. AVC is U-shaped because of _____.
17. Difference between ATC and AVC is_____.
18. Economic cost =_____+_____+_____.
19. Explicit cost is also known as _____.
20. TC starts from _____ because it has _____ in it which will never be_____.
21. Cost function =_____.
22. Per unit cost of production is known as_____.
23. Difference between total cost and total variable cost is_____.
24. MC crosses AC and AVC from_____.
25. TR =_____
26. AR =_____
27. MR =_____
28. In perfect competition AR =_____.
29. In imperfect competition AR _____ MR.
30. In perfect competition shape of AR is _____.

ANSWERS

1. Explicit cost	2. Implicit cost	3. Wages paid to the labourers, Rent for the land	4. Imputed rent of owners' occupied land, Salary of the owner working as manager
5. Variable cost	6. Fixed cost	7. Total Fixed cost	8. Total Variable Cost

9. Rectangular hyperbola	10. Marginal cost	11. Law of variable proportion	12. Rising
13. Falling	14. Minimum	15. Higher rate	16. Law of variable proportion.
17. AFC	18. Explicit cost, Implicit cost, Normal profit	19. Money cost	20. Above the origin, TFC, Zero
21. C = f(Q)	22. Average cost	23. Total fixed cost	24. Their minimum point
25. P x Q	26. TR/Q	27. TR_n- TR_{n-1}	28. MR
29. > (greater than)	30. A horizontal straight line.		

B. TRUE OR FALSE

1. Average cost can fall even when MC is rising.
Ans. [False] As MC curve cuts AC curve at its lowest point and after that MC curve rises because MC curve lies above it. True on the condition that MC will remain below AC.

2. As soon as MC starts rising AVC also starts rising.
Ans. [False] Because MC may start rising even when AVC is falling-

3. AC and AVC can be equal at any point of time.
Ans. [False] AC = AFC + AVC and AFC is never zero as TFC remains constant.

4. The difference between ATC and AVC is constant.
Ans. [False] Because ATC = AFC + AVC and AFC decreases with increase in the output.

5. As output increases, the difference between ATC and AVC falls and ultimately becomes zero.
Ans. [False] Because difference decreases with increase in the output but it will never become zero as TFC is never zero

6. AVC is a U-shaped curve.
Ans. [True] Because it follows law of variable proportions.

7. MC is not affected by fixed cost.
Ans. [True] As MC is the additional cost incurred on producing one additional unit of output and TFC remains fixed it does not change thus there is no effect of TFC on MC.

8. Both TC and TFC curves start from the origin.
Ans. [False] As both the curve start from Y-axis as both are same at zero level of output.

9. The marginal cost curve can intersect average cost curve at any point.
Ans. [False] As marginal cost curve can intersect the average cost curve only at its minimum point.

10. AFC curve touches the Y-axis because at a zero output, AFC is zero.
Ans. [False] AFC can never touch the Y-axis because at zero output, it is infinite.

11. All average cost curves are U-shaped.
Ans. [False] AC and AVC curves are U-shaped, but AFC is rectangular hyperbola.

12. Cost in economics means explicit cost only.
Ans. [False] As cost in economics means explicit and implicit cost both.

13. In case of constant prices, average revenue is more than marginal revenue.
Ans. [False] AR is equal to MR in case of constant prices.

14. Total revenue curve is a positively sloped straight line when price remains same at all levels of output.
Ans. [True] It happens because in case of constant prices, total revenue increases at a constant rate.

15. MR can neither be zero nor negative.
Ans. [False] MR can be zero and even negative, if price falls with rise in output.

16. TR is at its maximum point when MR is zero.
Ans. [True] TR rises as long as MR is positive and reaches at its maximum point when MR is zero.

17. AR curve always remain above MR.
Ans. [False] AR curve remain above MR curve only when price falls with rise in output. If price remains same then AR and MR curves coincide in a horizontal straight line parallel to the X-axis.

18. AR and price are one and the same thing.
Ans. [True] AR = TR/Quantity, TR = P x Q, thus AR = P x Q/Q = Price.

19. AR and MR curves slope downwards when more output can be sold by reducing the prices.
Ans. [True] AR falls with increase in sales at lower prices and revenue from every additional sales is less than AR as a result both AR and MR curves slope downwards from left to right.

20. Total revenue curve always starts from the origin.
Ans. [True] TR curve always starts from the origin as it zero at zero level of output.

C. MULTIPLE CHOICE QUESTIONS

COST OF PRODUCTION

1. Whatever expenditure is incurred for acquiring factor services is known as:
(a) social costs (b) opportunity costs
(c) money costs (d) real costs

2. What are those costs which firms incurred for obtaining factor services from outside?
(a) Production costs (b) Explicit costs
(c) Opportunity costs (d) Implicit costs

3. What are those costs for factors which are contributed by firm itself from its own sources?
(a) Implicit costs (b) Explicit costs
(c) Opportunity costs (d) Production costs

4. Costs forgone for the production of a good is known as:
(a) implicit costs (b) explicit costs
(c) opportunity costs (d) production costs

5. Costs of exertion, pains and efforts is known as:
(a) opportunity costs
(b) production costs
(c) real costs
(d) explicit costs

6. Per unit cost of production is known as
(a) cost production
(b) average cost
(c) marginal cost
(d) real costs

7. Cost of producing an additional unit of goods is known as:
(a) average costs
(b) marginal costs
(c) opportunity costs
(d) real costs

8. Those costs which do not change with the level of output is known as:
(a) fixed costs
(b) variable costs
(c) production costs
(d) implicit costs

9. Those cost which vary with the level of output are known as:
(a) fixed costs
(b) variable costs
(c) production costs
(d) explicit costs

10. Depreciation of machine interest on fixed capital, expenditure on building is an expenditure of which of following type?
(a) Variable costs
(b) Fixed costs
(c) Production costs
(d) Implicit costs

11. Expenditure on raw material, power etc., is of which of following nature?
(a) Fixed
(b) Variable
(c) Production costs
(d) Explicit

12. Write down the formula for calculating total cost.
(a) TC = AC x Q
(b) TC = TFC + TVC
(c) Both (a) and (b)
(d) Any of these

13. Write down the formula for total variable cost.
(a) TVC - ΣMC
(b) TVC - TC – TV
(c) TVC = AVC x Q
(d) All of these

14. How average cost is calculated?
(a) To
(b) AFC + AVC
(c) Both (a) and (b)
(d) None of these

15. How marginal cost is calculated?
(a) TVCn -TVCn - 1
(b) ΔTVC/ΔQ
(c) Both (a) and (b)
(d) None of these

16. MC curve crosses AC curve from:
(a) lowest bottom
(b) left to lowest bottom
(c) right of lowest bottom
(d) any of these

17. When AC rises, MC also rises:
(a) less fastly
(b) more fastly
(c) constant rule
(d) none of these

18.When AC falls, MC has a tendency of:
(a) falling less than AC
(b) falling sharply than AC
(c) remains constant
(d) none of these

19. Average fixed cost has a shape of:
(a) parallel to X-axis
(b) rectangular hyperbola
(c) downward sloping
(d) rising upward

20. Which of the following statement is true?
(a) AC = TFC - TVC
(b) AC = AFC + TVC
(c) AC = AFC + AVC
(d) None of these

21. Total cost of production of zero level of output is:
(a) zero
(b) positive
(c) negative
(d) none of these

22. AVC is .... of zero level of output.
(a) zero
(b) positive
(c) negative
(d) none of these

23. MC curve has the shape of:
(a) V-shape
(b) U-shape
(c) inverted U-shape
(d) none of these

24. Which of the following explains cost function?
(a) C = f(q)
(b) C = f(P)
(c) C = f(0)
(d) none of these

25. Difference between total cost and total variable cost is:
(a) TFC
(b) MC
(c) AC
(d) None of these

26. Difference between AC and AVC is of ... and it is declining.
(a) AFC
(b) MC
(c) total fixed cost
(d) None of these

27. Which of the following is a variable cost?
(a) Purchase of new machine
(b) Monthly expenditure of elasticity
(c) Manager's salary
(d) Purchase of raw material

28. If cost of producing 4 units is < 50 and that of 5 units is 68. Marginal cost would be:
(a) Rs. 17
(b) Rs. 18
(c) Rs. 12.5
(d) Rs. 13.6

29. Which of following statement is true?
(a) MC = AFC + AVC
(b) MC = TCN +1 - TCN-1
(c) AVC = TVC/Q
(d) None of these

ANSWERS

1.c	2.b	3.a	4.c	5.c	6.b	7.b	8.a	9.b	10.b
11.b	12.c	13.d	14.c	15.c	16.a	17.b	18.b	19.b	20.c
21.b	22.a	23.b	24.a	25.a	26.a	27.d	28.b	29.c	

CONCEPT OF REVENUE

1. What is price per unit is called?
(a) Average revenue
(b) Marginal revenue
(c) Total revenue
(d) Any of these

2. What is known as total income from sales?
(a) Income from sales
(b) Total revenue
(c) Average revenue
(d) Marginal revenue

3. What do we call as income from sale of an additional unit?
(a) Average revenue
(b) Marginal revenue
(c) Total cost
(c)

4. Which of the following is the formula for calculating average revenue (AR)?
(a) TR/Q
(b) AR x Q
(c) ∑ MR
(d) Q/TR

5. How total revenue is calculated?
(a) AR x Q
(b) ∑ MR
(c) Both (a) and (b)
(d) None of these

6. How marginal revenue is calculated?
(a) TRn - TRn - 1
(b) ΔTR/ΔQ
(c) Both (a) and (b)
(d) None of these

7. What is relationship between AR and MR under perfect competition?
(a) AR > MR
(b) AR < MR
(c) AR = MR
(d) AR ≠ MR

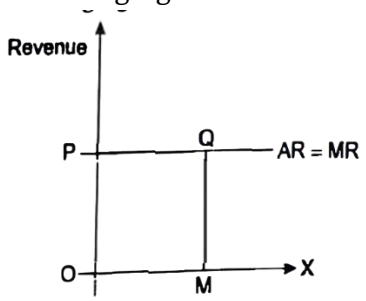
8. What is the shape of AR under perfect competition?
(a) Parallel to Y-axis
(b) Parallel to X-axis
(d) None of these
(c) Rising upward straight line

9. What is the shape of AR and MR under imperfect competition?
(a) Downward sloping
(b) Rising upward
(c) Both (a) and (b)
(d) None of these

10. How price is calculated from TR?
(a) Price = Total revenue/ Total sales
(b) Price = Total sales / Total revenue
(c) Both (a) and (b)
(d) None of these

11. Under imperfect competition when TR is maximum, MR would be:
(a) positive
(b) negative
(c) zero
(d) none of these

12. What can be expressed OPQM in the following figure?



(a) AR
(b) MR
(c) TR
(d) None of these

13. Which of the following statement is true?
(a) AR= MR/Q
(b) TR = MR x Q
(c) TR = ∑ MR
(d) All of these

14. Which of the following statement is true?
(a) AR =TR / Q
(b) AR = MR / TR
(c) AR = Q / TR
(d) AR = TRn - TRn - 1

15. Revenue of firm refers to:
(a) demand side for the firm
(b) supply side of the firm
(c) cost of the firm
(d) none of these

1.a	2.b	3.b	4.a	5.c	6.c	7.c	8.b	9.a	10.a
11.c	12.c	13.c	14.c	15.a					

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) :	When price remains the same at all levels of output, then the total revenue (TR) curve is a positively sloped straight line.
Reason (R) :	In the case of constant prices, MR is constant, i.e., TR increases at a constant rate.
2. Assertion (A) :	Total revenue is at its maximum point when marginal revenue is zero.
Reason (R) :	When every additional unit is sold at the same price, Marginal Revenue = Average Revenue.
3. Assertion (A) :	Total revenue (TR) curve always starts from the origin.
Reason (R) :	TR is 0 at zero level of output.
4. Assertion (A) :	When MR becomes negative, TR starts falling.
Reason (R) :	When price falls with rise in output, then MR is more than AR.
5. Assertion (A) :	Normal profit is a part of implicit cost.
Reason (R) :	Normal profit is the imputed value of entrepreneurial services provided by the owner.
6. Assertion (A) :	TFC curve is a vertical straight line parallel to the Y-axis.
Reason (R) :	TFC remains the same at all levels of output, even if the output is zero.
7. Assertion (A) :	Total variable cost (TVC) can be changed in the short-run.
Reason (R) :	TVC increases initially at a decreasing rate, then at a constant rate and finally, at an increasing rate.
8. Assertion (A) :	TC and TVC curves are parallel to each other.

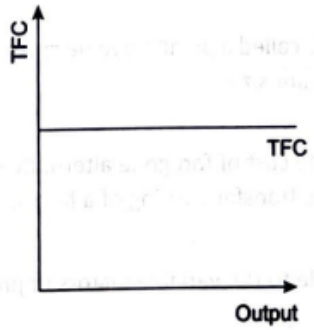
Reason (R) :	The vertical distance between TC and TVC curves is TFC, which remains constant at all levels of output.
9. Assertion (A) :	AFC curve can never touch the Y-axis.
Reason (R) :	TFC can never be zero.
10. Assertion (A) :	Short-run average cost curves are U-shaped
Reason (R) :	Short-run cost curves follow law of variable proportions.

ANSWERS

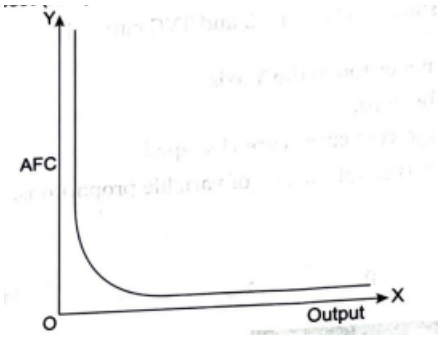
1.c	2.d	3.c	4.d	5.c	6.a	7.d	8.c	9.d	10.c

2 MARK QUESTIONS

1. What do you mean by costs?
Ans. By 'costs' in economics, we generally mean the price of the factors of production for producing a particular amount of output.
2. What do you mean by 'economic cost'?
Ans. By economic cost, we mean both explicit and implicit costs for producing a particular amount of output.
3. What is explicit cost?
Ans. The cost of production which can be measured in money terms, is known as explicit cost.
4. What is implicit cost?
Ans. The opportunity cost of producing a particular amount of output using some factors of production is called as the implicit cost.
5. What are the two components of short-run total cost?
Ans. The two components of short-run total cost are: (i) total fixed cost and (ii) total variable cost.
6. Define short-run TFC curve
Ans. The cost of production which remain fixed during short-run and do not vary with changes in the quantity of output, is called TFC.
7. Why does the short-run TFC curve become horizontal?
Ans. The short-run TFC curve becomes horizontal since the total fixed costs (TFC) remain fixed at all levels of output.



8. Define short-run variable costs.
Ans. The costs which arise during short-run due to the purchase of variable factors of production and the costs which vary with changes in the level of output are called as variable costs.
9. Define average fixed costs (AFC)
Ans. The AFC is the fixed costs per unit of production and it is derived through dividing the total fixed costs by the quantity of output.
10. What is the shape of the AFC curve?
Ans. The shape of the average fixed cost (AFC) curve is like a rectangular hyperbola.



11. Give two **Example**s of fixed costs of a firm.
Ans. The interest payment on loan and the salary payments for the permanent staff are the two **Example**s fixed costs of a firm during short-run.
12. Give two **Example**s of variable costs of a firm.
Ans. The costs of raw materials and the wage cost of casual workers are the two **Example**s of variable costs.
13. Define average variable costs (AVC).
Ans. The variable costs of production per unit of output, is called as the AVC.
14. What is average cost of production?
Ans. The total cost of production per unit of output is called as average cost (AC).
15. What is marginal cost (MC) of production?
Ans. The change in total cost of production due to a change in output by one unit is called as the MC of production.
16. What is the shape of a short-run TVC curve?
Ans. The short-run TVC curve looks like an 'inverted S'.
17. How can you derive the short-run TC curve from the TFC and TVC curves?
Ans. The short-run total cost (TC) curve can be derived from the vertical summation of total fixed cost (TFC) and total variable cost (TVC) curves.
18. Why does AFC fall with an increase in output?
Ans. The average fixed cost (AFC) falls with an increase in output (Q) since total fixed cost (TFC = AFC x Q) remains constant at all levels of output.
19. What is plant curve?
Ans. The short-run average cost (SAC) curve is called a plant curve because each SAC curve corresponds to a given plant size. During short-run, a firm cannot change this plant size.
20. What is opportunity cost?
Ans. Opportunity cost of any commodity is the cost of foregone alternative, i.e., the next best alternative which is sacrifice to produce this commodity. It is also known as transfer earning of a factor.
21. What is variable cost?
Ans. Variable cost refers to the payment made to the variable factors of production. It is also known as prime cost or dire cost. It varies with change in output.

22. Explain the term average cost.
Ans. Average cost is the cost per unit of output. It is calculated by dividing the total cost by the number of units produced.
23. What does the average revenue signify?
Ans. Average revenue signifies the revenue earned by a firm per unit of output sold. It is estimated by dividing total revenue by the quantity of output sold in the market. It also signifies the demand in the market.
24. What is meant by total revenue?
Ans. Total revenue refers to the total amount of money received by the firm from the sale of a given quantity of output. It is he product of price and quantity, i.e., TR = P x Q.
25. Give the meaning of marginal revenue.
Ans. Marginal revenue is the addition to the total revenue when one extra unit of output is sold in the market.
26. Which concept of revenue is equal to price?
Ans. AR is always equal to price. Since, AR = TR /Q = P x Q/Q =P
Therefore, AR = Price.

3-4 MARKS QUESTIONS

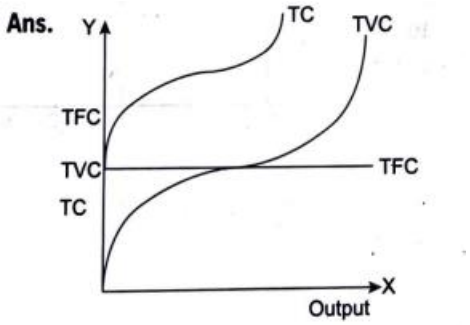
1. What is the reason behind the "U'-shaped of the AVC curve during the short-run?
Ans. The reason behind the U-shaped short-run average variable cost (AVC) curve can be explained by the law of returns to a variable factor. It shows the relationship between change in output of a firm and the change in any variable factor of production (say, labour) during the short-run. At the initial stage, there are increasing returns to a variable factor. So, AVC declines with an increase in output. At the next stage there are constant returns to a variable factor. So, AVC remains unchanged at the minimum level. Finally there are decreasing returns to the variable factor. So, the AVC rises with an increase in output. Hence, AVC curve becomes U-shaped

2. Indicate the relationship between average cost and marginal cost of production.
Ans. The cost of production per unit of output of a firm is called the average cost (AC) of production. The change in total costs of production due to a small change in output, is called the marginal cost (MC) of production. The relationship between AC and MC is shown below:
(i) When AC falls, MC < AC.
(ii) When AC is minimum, MC = AC.
(iii) When AC rises, MC > AC.

3. What are the elements of the total cost of production in the short-run ?
Ans. During the short-run, total cost (TC) of production includes, both total fixed costs (TFC) and total variable cost (TVC) of production of any firm.
TC = TFC + TVC during the short-run for **Example**, fixed salary paid to the managers and security guards, fixed rent paid for the factory shed, fixed amount of interest payment on loan, etc., are fixed costs. These costs remain, unchanged at all levels of output. However, TVC varies with a variation in the level of output. For instance, wages paid to the causal workers, costs of raw materials, etc., are the variable costs of a firm.

4. Draw TFC, TVC and TC curves in a single diagram.

Ans.



5. Why is the MC curve in the short-run U-shaped ?

Ans. MC curve is U-shaped because of law of variable proportions. MC falls in the beginning then reaches at minimum becomes constant for a while and ultimately it rises. Initially, production is subject to law of increasing returns then diminishing returns and ultimately to the negative returns, i.e., why AC is U-shaped.

6. Why is AC curve in the short-run U-shaped?

Ans. AC curve is U-shaped because of law of variable proportions. AC falls in the beginning then reaches at minimum becomes constant for a while and ultimately it rises. Initially, production is subject to law of increasing returns then diminishing returns i.e., why MC is U-shaped.

7. What is the reason behind the "U'-shape of
AVC curve during the short-run?

Ans. The reason behind the U-shaped short -run average variable cost (AVC) curve can be explained by the law of returns to a variable factor. It shows the relationship between change in output of a firm and the change in any variable factor of production (say, labour) during the short-run. At the initial stage, there are increasing returns to a variable factor. So, AVC declines with an increase in output. At the next stage there are diminishing returns to a variable factor. So, AVC remains very low at the minimum level. Finally there are negative returns to the variable factor. So, the AVC rises with an increase in output. Hence, AVC curve becomes U-shaped.

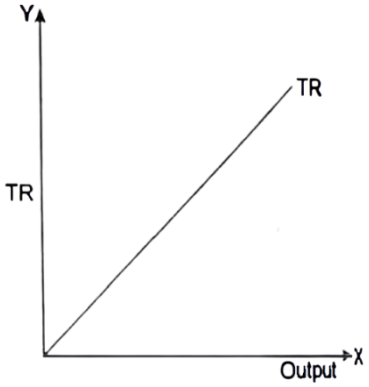
8. Which cost, fixed or variable, determines marginal cost? Give reasons.

Ans. Variable cost determines marginal cost. The reason is that the fixed cost do not change with change in level of output in the short-run. It is the variable cost only which changes with level of output. MC by definition is the addition made to total cost due to an additional unit of commodity produced and this addition in TC is due to variable cost only as fixed cost remains constant. Therefore, MC depends on variable cost.

9. Why is total revenue curve under perfect competition is a straight line passing through the origin?

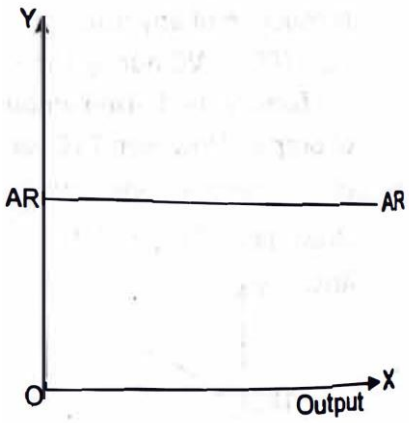
Ans. Under perfect competition, each firm is a price taker and cannot make any change in the price of the commodity.

Quantity	Price	TR=Price x Quantity
1	10	10
2	10	20
3	10	30



10. Why is AR always equal to MR in a perfect competitive firm?

Ans. AR is the total revenue per unit of output, MR is the additional revenue earned by
selling one additional unit of output. Since, in a competitive market price remains fixed
and firm is a price taker then the per unit revenue and extra revenue both become the
same. Thus, in a perfect competition market P = AR = MR.



Quantity	Price	TR= Price x Quantity	AR = TR/Q	MR=TRn-TRn-1
1	10	10	10	10
2	10	20	10	10
3	10	30	10	10

6-8 MARKS QUESTIONS

1. What is the relationship between TC, TVC and TFC?

Ans. Total cost of a firm is the sum of its total variable costs and total fixed cost

TC =TFC + TVC

Fixed costs are costs which do not vary with output and cannot be altered in the short-run. Total fixed costs remain constant when the output is increased. Therefore, total fixed cost curve is a straight line parallel to the x-axis.

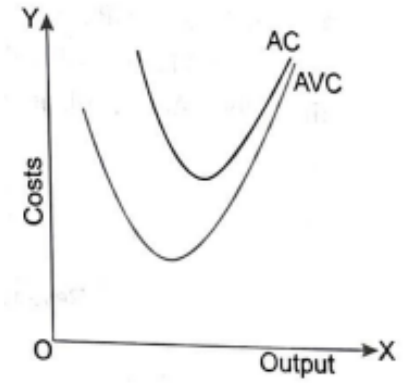
Total variable costs vary with the level of output. TVC is zero when nothing is being produced. It is positively sloped starting from zero. In the initial stages it rises a diminishing rate and then it starts rising at an increasing rate. Total cost curve can be obtained by adding vertically TFC curve and TVC curve because total cost is the sum of total fixed cost and total variable cost. The shape of the TC curve is exactly the same as that of total variable cost (TVC) curve.

2.With increase in output, the ATC and AVC curves come closer and closer. Why? Can they meet? Give reasons in support of your answer.

Ans. Average fixed cost is the fixed cost per unit of output. Since, the total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases. Therefore, AFC curve in the beginning slopes downward and when output becomes very large AFC approaches zero.

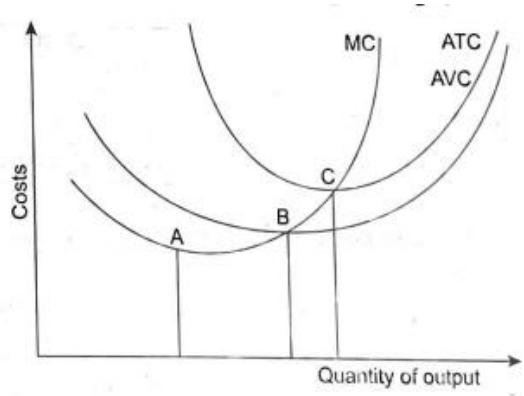
Average variable cost is total variable cost divided by the number of unit of output. AVC curve will fall initially because of the occurrence of increasing returns but beyond the normal capacity output, variable costs would increase because of diminishing returns.

Average total cost is the sum of AFC and ATC. As output increases AFC keeps on becoming
smaller and smaller. Therefore, the vertical distance between ATC and AVC keeps on becoming less and less. When AFC approaches zero. ATC approaches AVC. Therefore, the ATC seems to be coming closer and closer to AVC.



The two curves would seem to meet but they would never meet. ATC = AVC + AFC. If the two curves meet it implies at that level of output ATC = AVC or AFC is zero but AFC can never be zero though it may become extremely small when output becomes quite large.

3. Why does MC curve intersect the AVC and AC curves at their lowest point ? Explain it with the help of diagram.

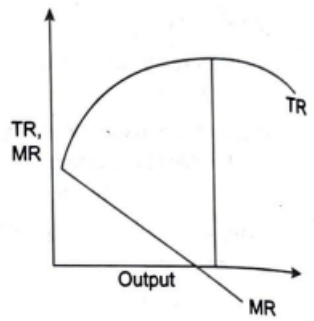


Ans. We know that as long as MC is less than AVC or AC, the two curves cannot rise. The AVC curve falls till the point B in the diagram because MC lies below it. At the point of intersection, the AVC is equal to MC but after point B, MC rises and becomes greater than AVC and therefore, AVC gets pulled up making B automatically therefore minimum point of the AVC curve.
Similarly, till point C we find MC less than ATC therefore ATC keeps falling but beyond this point MC becomes greater than ATC pulling ATC upwards and making point C, its minimum point. Therefore, we find MC always intersecting AVC and ATC at their minimum points.

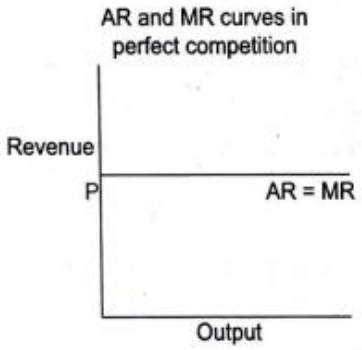
4. Classify the following into fixed cost and variable costs. Give reasons:
(i) Rent for a land.
(ii) Minimum telephone bill.
(iii) Cost of raw material.
(iv) Wages to permanent staff.
(v) Interest on capital.
(vi) Payment for transportation of goods.
(vii) Daily wages.
Ans. (i) Rent for a shed is fixed cost because it is a contractual payment fixed in advance and it does not change with output
(ii) Minimum telephone bill is fixed cost because this payment has to be made irrespective of the output produced.
(iii) Cost of raw material is a variable cost as it increases/decreases with increase/decrease in the level of output produced
(iv) Wages to permanent staff is fixed cost as permanent staff will have to be paid the same salary whatever may be the output level being produced by the firm.
(v) Interest on capita is fixed cost because the firm owner has borrowed and invested the sum and now, whatever may be the level of output produced, interest on borrowed sum be, it will have to be paid.
(vi) Payment for transportation of goods is variable cost, because if the output increases, more would have to be spent on the transportation of this higher output.
(vii) Daily wages is variable cost because for producing a higher output, one will have to employ more workers on daily wages and hence this payment would increase with output.

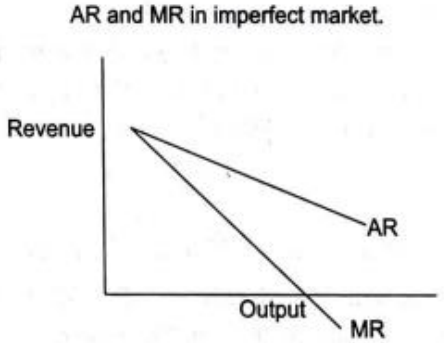
5. Explain the relationship between TR and MR.
Ans. Relationship between TR and MR:
(i) As long as TR increases at a diminishing rate, MR decrease remains positive.

(ii) When TR becomes maximum and constant, MR becomes zero.
(iii) When TR starts decreasing MR will go into negative.



6. Explain the relationship between AR and MR.
Ans. (i) AR and MR both are calculated from TR. AR = TR/Q; MR = TRn - Rn-1
(ii) If AR is falling MR will also fall but at a higher rate.
(iii) When AR is horizontal straight line then MR will also be horizontal straight line.





QUESTIONS BASED ON DIFFERENTIATION

1. Differentiate between short-run costs and long-run costs.
Ans.

S.No.	Basis	Short-run Costs	Long-run Costs
1	**Meaning**	Short-run costs consists of both fixed costs as well as variable costs.	Long-run costs refer to the variable costs only.
2	**Shape of the curve**	Short-run cost curves are of U-shaped because of the operation of the law of variable proportions.	LAC and LMC are U-shaped because of the operation of the law of returns to scale.
3	**Other name**	The SAC curve is a plant curve, as it corresponds to a given plant size.	The LAC curve is an envelope curve, as it envelopes so many SAC curves.
4	**The minimum point**	The minimum point of SAC indicates the optimum utilisation of a particular plant size.	The minimum point of LAC indicates the optimum plant size among various sizes.

2. Differentiate between accounting and opportunity cost.

Ans.

S.No.	Basis	Accounting Cost	Opportunity Cost
1	**Meaning**	It refers to money expenses which the firm has to incur in purchasing or hiring the factor services.	It refers to the next best alternative good i.e., given up to produce a good.
2	**Books of accounts**	It plays very important role in books of accounts because they are recorded as monetary expenses made by firms for producing goods and services.	These are not included in books of accounts.
3	**Other name**	Known as money cost or business cost also.	known as economic cost also.

3. Differentiate between explicit cost and implicit cost.

Ans.

S.No.	Basis	Explicit Cost	Implicit Cost
1	**Meaning**	These are the costs incurred on purchasing or hiring inputs for producing goods and services.	These are the costs incurred on those inputs which are owned by the owner himself.
2	**Alternative name**	These are known as money cost as they are paid in terms of money. These are known as accounting cost also as they are included in the books of accounts to maintain records.	These are known as imputed or estimated costs also.
3	**Examples**	(i) Payment of wages to labourers. (ii) Payment for rent. (iii) Payment for raw material used in production.	(i) Rent on entrepreneur's own land. (ii) Interest on owner's own capital. (iii) Normal profit.

NUMERICALS

Important Formulae

1. TC = TFC + TVC or AC x Q
2. TFC = TC - TVC or AFC x Q or TC at 0 output
3. TVC = TC - TFC or AVC x output (Q)
4. AVC = TVC/Q or AC - AFC
5. AFC = TFC/Q or AC - AVC
6. AC = AFC + AVC or TC/Q
7. MC = TCn - TCn -1 or TVCn - TVCn - 1
8. TR = P x Q or AR x Q or EMR
9. AR = TR / Q or = price
10. MR = **Δ**TR / **Δ**Q or TRn - RTn -1

1. The cost function of a firm is given below: calculate TFC, TVC, AFC, AVC, AC and MC. **(6 Marks)**

Outputs(Units)	0	1	2	3	4	5
Totals Cost Rs	50	90	120	135	138	145

Ans.

Q (Units)	TC (Rs)	TFC (Rs)	TVC (Rs)	AFC (Rs)	AVC (Rs)	AC (Rs)
0	50	50	-	-	-	-
1	90	50	4	50	40	90
2	120	50	70	25	35	60
3	135	50	85	16.67	28.33	45
4	138	50	88	12.5	22	34.5
5	145	50	95	10	19	29
6	190	50	140	8.33	23.33	31.66

2. Calculate TC, AC, AFC, AVC and MC on the basis of the following information: **(6 Marks)**

Output	0	1	2	3	4
Total Fixed Cost (Rs)	60	60	60	60	60
Total Variable Cost (Rs)	0	50	90	180	300

Ans.

Output	0	1	2	3	4
Total Fixed Cost (7)	60	60	60	60	60
Total Variable Cost (7)	0	50	90	180	300
Total Cost	60	110	150	240	360
Average Cost	-	110	75	20	90
Average Fixed Cost	-	60	30	82	15
Average Variable Cost	-	50	45	60	75
Marginal Cost	-	50	40	90	12

3. Complete the following table: **(6 Marks)**

Output	TVC (Rs)	AVC (Rs)	MC (Rs)
1	10	-	-
-	-	8	6
3	27	9	-
-	-	10	13

Ans.

Output	TVC (Rs)	AVC (Rs)	MC (Rs)
1	10	10	-
2	16	8	6
3	27	9	11
4	40	10	13

4. Calculate the TVC and MC from the following table when total fixed cost is Rs 12. **(4 Marks)**

Output (Units)	1	2	3	4
TC (Rs)	20	26	31	38

Ans.

Output (Units)	1	2	3	4
TC (Rs)	20	26	31	38
TVC	8	14	19	26
MC	-	6	5	7

5. Calculate TVC and TC from the following cost schedule of a firm whose fixed cost is Rs 10. **(4 Marks)**

Output (units)	1	2	3	4
MC (Rs)	6	5	4	6

Ans.

Output (units)	1	2	3	4
MC (Rs)	6	5	4	6
TFC	10	10	10	10
TVC	6	11	15	21
TC	16	21	25	31

6. Calculate TC and AVC of a firm at each level of output from the following cost schedule. **(4 Marks)**

Output(units)	1	2	3	4	5	6
AFC (₹)	60	30	20	15	12	10
MC (₹)	32	30	28	30	35	43

Ans.

Output(units)	1	2	3	4	5	6
AFC (₹)	60	30	20	15	12	10
MC (₹)	32	30	28	30	35	43
TFC	60	60	60	60	60	60
TVC	32	62	90	120	155	198
TC	92	122	150	180	215	258
AVC	32	31	30	30	31	33

7. From the following data given below: (i) Marginal cost and (ii) Average cost. **(3 Marks)**

Output	1	2	3	4	5	6	7	8
Total Cost (₹)	10	20	28	34	38	42	48	56

Ans. (i) MC = -, 10, 8, 6, 4, 4, 6, 8 and (ii) AC = 10, 10, 9.3, 8.5,7.6, 7, 6.

8. Find total fixed cost, average fixed cost, total variable cost and marginal cost from the following data: **(4 Marks)**

Unit of output	0	1	2
Total Cost (₹)	75	96	110

Ans. TFC = ₹ 75, 75 and 75; AFC = ₹ 75.0, 37.5; TVC = ₹ 0, 20, 35; MC= ₹ -, 20, 15

9. Complete the following table:

Output	Total Cost (₹)	Marginal Cost (₹)	Average Variable Cost (₹)
0	60	-	-
1	140	-	-
2	190	-	-
3	240	-	-
4	300	-	-

Ans. MC = ₹ -, 80, 50, 50 and 60; AVC = ₹ -, 80, 65, 60 and 60.

10. Calculate total variable cost and marginal cost from the following cost schedule of a firm whose total fixed costs are ₹12. **(3 Marks)**

Output Units	Total Cost (₹)
1	20
2	26
3	31
4	38

Ans. TVC = ₹ 8, 14, 19, 26; MC = ₹ 8, 6, 5, 7.
[Hint: At MC (Ist unit of output) = TC - TFC (= TVC) at unit 1 of output MC = TVC]

11. Total Fixed Cost is ₹ 90. Complete the following Table: **(3 Marks)**

Unit of Output	Marginal Cost (₹)	Total Cost (₹)	Average Total Cost (₹)
1	10	-	-
2	20	-	-
3	15	-	-

Ans. TC=₹ 100, 120 and 135; ATC = ₹ 100, 60 and 45.

12. Calculate total variable cost and marginal cost at each given level of output from the following table: **(3 Marks)**

Output (Units)	**0**	**1**	**2**	**3**	**4**
Total Cost (in ₹)	**40**	**60**	**78**	**97**	**124**

Ans. TVC = ₹ 0, 20, 38, 57 and 84; MC = ₹ -, 20, 18, 19 and 27.

13. Complete the following table: **(3 Marks)**

Output(Units)	TC (₹)	AVC(₹)	MC(₹)
0	80	-	-
1	180	-	-
2	270	-	-
3	350	-	-
4	440	-	-

14. Complete the following table:

Output(Units)	TC (₹)	AVC(₹)	MC(₹)
0	50	-	-
1	80	-	-
2	100	-	-
3	125	-	-
4	158	-	-

Ans. AVC = ₹ 30, 25, 25 and 27, MC = ₹ 30, 20, 25 and 33.

15. The total fixed cost of a firm is ₹ 12. Given below is its marginal cost schedule. Calculate total cost and average variable cost for each level of output. **(4 Marks)**

Output (Units)	**1**	**2**	**3**	**4**	**5**	**6**
Marginal Cost (in ₹)	**9**	**7**	**2**	**4**	**8**	**12**

Ans. TC = ₹ 21, 28, 30, 34, 42, 54; AVC = 9, 8, 6, 5.5, 6, 7.

16. A firm is fixed cost is ₹ 2,000. Compute the TVC, AVC, TC and ATC from the following table: **(6 Marks)**

Output (Units)	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Marginal Cost (in ₹)	**2000**	**1500**	**1200**	**1500**	**2000**	**2700**	**3500**

Ans. TVC = ₹ 2000, 3500, 4700, 6200, 8200, 10900, 14400;
AVC = ₹ 2000, 1750, 1567, 1550, 1640, 1816.6, 2057;
TC = ₹ 4000, 5000, 6700, 8200, 10200, 12900, 16400;
ATC = ₹ 4000, 2750, 2233,2050, 2040, 2150, 2343.

17. From the following cost schedule of a firm, calculate marginal cost and average variable cost of each level of output Fixed cost is ₹ 120. **(3 Marks)**

Output (Units)	**1**	**2**	**3**
Average Variable Cost (in ₹)	**40**	**32**	**36**

Ans. MC = ₹ 40, 24, 44; ATC =₹ 160, 92, 76

18. Given below is the cost schedule of a firm. Its total fixed costs are < 90. Calculate marginal cost and total cost at each level of output. **(4 Marks)**

Output (Units)	**1**	**2**	**3**
Average Variable Cost (in ₹)	**30**	**24**	**27**

Ans. MC = ₹30, 18, 33; C= ₹120, 138, 171.

19. Complete the following table: **(4 Marks)**

Output(Units)	**TVC (₹)**	**AVC(₹)**	**MC(₹)**
1	-	12	-
2	20	-	-
-	-	10	10
4	40	-	-

Ans. Output 3; TVC = ₹ 12, 30; AVC = ₹ 10, 10; MC = ₹ 12, 8, 10.

20. Complete the following data: **(3 Marks)**

Output(Units)	**AFC (₹)**	**MC(₹)**	**TC(₹)**
1	-	-	-
2	-	10	82
3	20	8	-
4	40	-	99
5	12	10	-

Ans. AFC = ₹ 60, 30 and 15; MC = ₹ 12 and 9; TC= ₹ 72, 90 and 109.

21. Calculate TR, AR and MR from the following data: **(6 Marks)**

Price Per Unit (₹)	**10**	**9**	**8**	**7**	**6**	**5**	**4**
Unit Sold	**10**	**20**	**30**	**40**	**50**	**60**	**70**

Ans. TR = ₹ 100, 180, 240, 280, 300, 300 and 280; AR = ₹ 10, 9, 8, 7, 6, 5 and 4; MR = ₹ 10, 8, 6, 4, 2, 0 and - 2.

22. Find out the average revenue and marginal revenue on the basis of following data: **(3 Marks)**

Unit of Production (₹)	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Unit Sold	**20**	**36**	**48**	**56**	**60**	**60**	**56**

Ans. AR = ₹ 20, 18, 16, 14, 12, 10 and 8; MR = ₹ 20, 16, 12, 8, 4, 0 and - 4.

23. Complete the following table: **(3 Marks)**

Units	TR	AR	MR
1	-	10	-
2	-	9	-
3	-	9	-

Ans. TR = ₹ 10, 18 and 27; MR = ₹ 10, 8 and 9.

24. Complete the following table: **(3 Marks)**

Units Sold	TR	MR	AR
1	10	10	-
2	-	-	9
3	24	-	-
4	-	4	7
5	30	-	6
6	30	0	-
7	28	-	4
8	-	-4	3

Ans. TR =₹ 18, 28 and 24; MR = ₹ 8, 6, 2 and -2; AR = ₹ 10, 8 and 5.

25. Complete the following table: **(4 Marks)**

Output(Units)	Price(₹)	TR(₹)	MR(₹)
4	9	36	-
5	-	-	4
6	-	42	-
7	6	-	-
8	-	40	-

Ans. Price = ₹ 8, 7 and 5; TR = ₹ 40, 42; MR = ₹ 2, 0 and -2.

CASE BASED QUESTIONS

CASE 1

Ramesh wants to start his own salon. He has an extra room in his house, and he wants to utilise it for his business. He himself will be working as a hair stylist but he must hire some workers also. He purchased furniture for his salon and got it decorated also. In the beginning of his business the revenue earned by him was negligible but cost of production was surely too much.

Answer the following questions:

1. What do you mean by explicit cost? Which is explicit cost in the above paragraph?
2. What is implicit cost in the above paragraph and why?
3. What do you call the cost which has to be incurred without the production also?
4. What is economic cost?

Ans. 1. All cash payments made by the firm to outsiders for their services and goods are known as explicit costs. Purchasing furniture and expenditure on decoration of the shop are included in explicit cost.
2. Shop, owner himself working as designer implicit cost as implicit costs are the costs of self-owned resources.
3. Fixed cost.
4. Economic cost = Explicit cost + Implicit cost + Normal profit.

CASE 2

Production is done in two time periods, short run and long run. Short run is a time when all factors of production can not be changed to change the output. It is the time when most of the average cost curves are U shaped. Long run is the time when all factors of production can be changed.

Answer the following questions:

1. What do you mean by factors of production?
2. What do you call that cost which can not be changed in the short run? Give **Example**s.
3. What do you mean by variable cost? Give **Example**s.
4. Why average cost curves in the short run are U shaped?

Ans. 1. Factors of production are resources that are the building blocks of the economy; they are what people use to produce goods and services. Economists divide the factors of production into four categories: land, labor, capital, and entrepreneurship.
2. Fixed cost. **Example** : Rent paid for land.
3. Expenditure incurred on variable factors of production is known as variable cost . Like wages paid to the labourers.
4. They are U shaped as they follow law of variable proportion.

QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. Do fixed costs affect marginal costs?

Ans. Fixed costs do not affect the marginal costs as there is no change in total fixed cost as the level of output increases in the short-run.

2. Show that the AVC curve is the mirror image of the average product (AP) curve of the variable factor.

Or Show the relation between AVC and AP.

Ans. We know that the average variable cost (AVC) is the total variable cost (TVC) per unit of output (Q). Let us assume that labour (L) is the only variable factor and the average wage rate (w) is given. So, TVC = W.L

$$AVC = TVC/Q = W.L/Q = w/Q/L = w/ AP$$

Where AP = average product of labour.
Now, according to the law of returns to a variable factor, AP increases at first with the increase in labour employment. Greater labour employment means great amount of output.

Thus, at the initial stage as AP rises, AVC will fall with an increase in output (since w is constant). When AP becomes maximum, then AVC is minimum. Then, as AP falls, AVC rises.

3. How can you give an estimate of TVC from the MC schedule ?
Ans. The marginal cost (MC) of production depends upon the change in total cost (TC) due to a change in output by one unit. But during short-run, the change in TC depends only on the change in total variable cost (TVC) since fixed cost remains unchanged. Hence, the TVC for producing a given quantity of output (Q) can be estimated from the summation of MC for different amount of output up to that level.

TCC = MC.

Let us consider the following cost schedule:

Q(Units)	TFC(₹)	TVC(₹)	MC(₹)	MC (₹)
0	10	0	10	-
1	10	5	15	5
2	10	9	19	4
3	10	12	22	3
4	10	14	21	2

4. What is the relationship between average cost, average fixed cost and average variable cost ?
Ans. During the short-run, the total cost (TC) of production of a firm consists of both total fixed costs (TFC) and total variable costs (TVC). The cost per unit of output is called the average cost (AC). Similarly, the fixed costs and variable costs per unit of output are termed as average fixed cost (AFC) and average variable cost (AVC) respectively.
Therefore, during the short-run, AC = AFC + AVC
[TC= TFC + TVC or TC/ Q= TFC/Q + TVC/Q or, AC = AFC + AVC; Where Q = Output].

5. Is it ever possible for the short-run total cost curve to start from the origin?
Ans. Generally, the total variable cost (TVC) curve starts from the origin because TVC = 0 when the level of output (Q) = 0. It takes the shape of inverted 'S' because of the operation of the law of returns to a variable factor. However, the short-run total cost curve cannot start from the origin because it includes some fixed costs (which are to be Incurred by a firm even when its output is zero).

6. What will be the relation between AC and MC when AC is at its minimum?
Ans. The average cost (AC) will be equal to the marginal cost (MC)

7. If total output is 'zero'. what will be the short-run total cost?
Ans. At zero output level, the short-run total cost will be equal to the total fixed costs.

8. What will be the relation between AC and MC when AC curve is downward sloping (or, when AC starts falling)?
Ans. When AC curve is downward sloping (i.e., when AC is falling) then MC becomes less than AC.

9. Why does the MC curve cut the AC curve from below at the minimum point of the AC curve?
Ans. This happens because when marginal cost (MC) is rising then it becomes equal to the average cost (AC) only when AC is minimum.

10. 'Marginal cost is influenced only by the variable cost'. Give reasons.
Ans. During short-run, total cost changes due to the change in variable cost only (since fixed cost does not change) and therefore, the marginal cost is influenced only by the variable cost.
11. 'All short-run average cost curve are U-shaped'. Do you agree? If not, why?
Ans. I do not agree with this statement. Although the short-run average cost (AC) and average variable cost (AVC) curves are U-shaped, the average fixed cost (AFC) curve is not U-shaped. It takes the shape of a rectangular hyperbola

12. How would you draw the average fixed cost curve of a firm?
Ans. The average fixed cost (AFC) of production of any firm indicates the fixed cost per unit of output during the short-run Since, total fixed cost (TFC) remains unchanged, so AFC declines continuously with an increase in the level of output (Q) However AFC x Q = TFC. Therefore, the AFC curve will be negatively sloped and take the shape of a rectangular hyperbola (graph).

13. Explain the concept of cost function.
Ans. The functional relationship between cost and output produced is known as cost function. Symbolically, C= f(Q), where Q = Units of output, C= cost.

14. Do ATC and AVC curves intersect? Give reason.
Ans. No, AVC never intersects ATC because AFC which is present in the gap between ATC and AVC reduces with increase in the level of output but it will never become zero as TFC is always positive.

15. Can the average cost be less than the marginal cost when average cost is rising?
Ans. Yes.

16. What is the difference between TC and TVC called?
Ans. The difference between TC and 1VC is called TFC.

17. At what point does the MC curve cuts AC curve?
Ans. At the minimum point of AC or ATC curve.

18. Can there be some fixed cost in the long run? If not why?
Ans. No, there cannot be any fixed cost in the long run. The reason is that in the long run all inputs can change.

19. When the total cost curve is horizontal how much is the marginal cost?
Ans. When the total cost curve is horizontal it means that the total cost does not change with the change in the output therefore, marginal cost will be zero because MC = TCn, - TC-n-1.

WORKSHEET FOR SELF EVALUATION

1. Why does the difference between average cost and average variable cost decreases with increase in the level of output? **(2 Marks)**
2. Draw AVC, AC and MC curves in a single diagram. **(3 Marks)**
3. Classify the following into fixed cost and variable cost: **(6 Marks)**
   - Expenditure on power and fuel.
   - Minimum electricity bill
   - Wages to permanent staff.
   - Daily wages.
   - Interest on capital.
   - Payment for transportation of goods.
   - Telephone charges beyond the minimum.
   - Rent for a building.
4. Why does the minimum point of AC curve fall towards right of AVC curve? **(2 Marks)**
5. MC can be calculated the from TC and TVC and is not affected by TFC. Why? **(4 Marks)**
6. Why does the MC curve cut the AVC curve at the minimum point of AVC curve?
7. Why is AR always equal to MR for a competitive firm? **(2 Marks)**
8. What is explicit cost? Are the following included in it? **(4 Marks)**

(i) Payment of wages to employees of the firm.
(ii) Wages for entrepreneur's own labour.
(iii) Rent of entrepreneur's own building.
(iv) Payment of electricity bill.

9. Draw TR, AR and MR curves of a market which decreases price of its products to sell more. **(6 Marks)**
10. AR = P, why, prove it. **(2 Marks)**
11. How will MR behave if AR is constant? **(2 Marks)**
12. Differentiate between explicit and implicit cost with the help of an example. **(4 marks)**
13. The cost function of a firm is given below : Calculate TFC, TVC, AFC, AVC, AC and MC. **(6 Marks)**

Output (Units)	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Total Cost (in ₹)	**50**	**90**	**120**	**135**	**138**	**145**	**190**

14. Complete the following table: **(4 Marks)**

Output(Units)	TVC (₹)	AVC(₹)	MC(₹)
1	10	-	-
-	-	8	6
3	27	10	-
-	-	10	13

15. Calculate TVC and TC from the following cost schedule of a firm whose fixed cost is ₹ 10. **(3 Marks)**

Output (Units)	**1**	**2**	**3**	**4**
MC (₹)	**6**	**5**	**4**	**6**

16. Calculate TC and AVC of a firm at each level of output from the following cost schedule: **(4 Marks)**

Output (Units)	1	2	3	4	5	6
AFC(₹)	60	30	20	15	12	10
MC (₹)	32	30	28	30	35	43

17. With the help of the given data, calculate the values of TR, AR and MR: **(3 Marks)**

Price (₹)	10	20	30	40	50	60	70
Units Sold	7	6	5	4	3	2	1

18. Complete the following table: **(3 Marks)**

Price (₹)	10	11	12	13	14	15	16
Output(₹)	-	9	-	7	-	5	-
MC (₹)	100	-	96	-	84	-	64